

# Class - 14

# INTRODUCTION

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HYPOTHESIS TESTING

# Hypothesis Testing

Hypothesis testing is like a **decision-making framework** in statistics.

We take data from a **sample**, then test if what we see is really true for the **whole population**.

In simple words:

It's like being a judge, we look at evidence (sample data) and decide if we should believe a claim about the population.

# Hypothesis Testing

There are two type of Hypothesis -

## 1. Null Hypothesis ( $H_0$ ) :

The “no difference” or “no effect” statement.

It assumes nothing unusual is happening.

## 2. Alternative Hypothesis ( $H_1$ ) :

Competes with  $H_0$ , and represents a change, an effect, or a difference.

In hypothesis testing, we always **assume  $H_0$  is true first**. Then, based on the evidence, we decide whether to keep it or reject it.



# Hypothesis Testing

Let's see an example for Hypothesis Testing,

"Let's say I have a coin, and I want to test whether the coin is fair or not by flipping it 100 times."

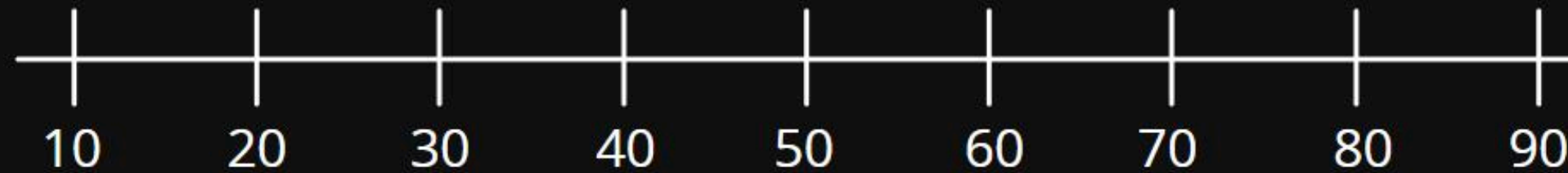
- 1) Null hypothesis - Coin is Fair.
- 2) Alternate Hypothesis - Coin is unfair.
- 3) now we will do the experiment
- 4) after Experimenting we will Accept or reject the null hypothesis.

# Hypothesis Testing

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Experiment -

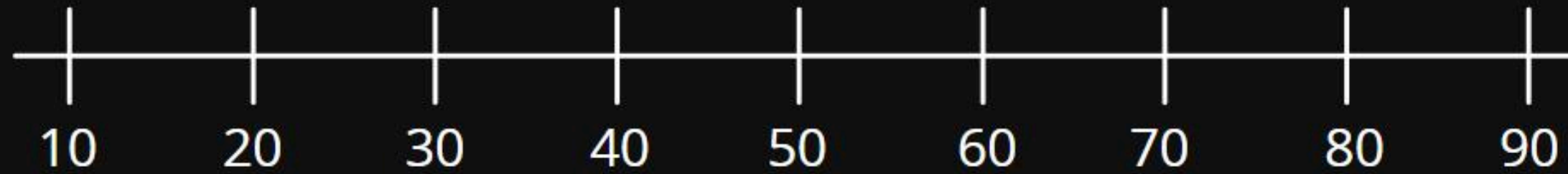
Let's say before experiment we got the information that our mean is 50 and standard deviation is 10. and we have a normal distribution.



Now I performed the experiment and I got head 30 times so can we say our coin is fair.

So to see the whether the coin is fair or not our experiment head count must be close to mean but how much close for this we will use a important concept of significance value.

# Hypothesis Testing



Significance value is set by the domain experts and let's say the significance value is 0.05 and it is denoted by  $\alpha$  and using this significance value we calculate the confidence interval and that will be,

$$1 - 0.05 = 0.95$$